

Assignment: Extend the AI Email Assistant with Gmail Reading and Reply Capabilities

Note: Before starting assignment read the following instructions carefully

Important Note

If you have already configured Google OAuth credentials for the **Gmail Send Email** node:

1. You do NOT need to create a new Google Cloud Project.
2. You do NOT need to create new OAuth credentials.
3. You can reuse the same Gmail credential for:

- **Send Email**
 - **Read Email**
 - **Reply Email**
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Objective

Enhance the AI-powered Email Assistant developed in *class* by adding **Gmail reading and email reply** functionality.

The current implementation can:

- Receive user requests through a Supervisor Agent
- Use SerpAPI for web searches
- Execute an Email Workflow
- Send emails using the Gmail Send Email tool

Your task is to extend the system so the AI can also:

1. **Read emails from Gmail**
 2. **Reply to existing emails**
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Requirements

Part 1: Read Gmail Tool

Create a new tool connected to the Email Agent.

Functional Requirements

The AI should be able to handle requests such as:

- Read my latest email
- Show unread emails
- What emails did I receive today?
- Read the email from Ali
- Summarize my latest email

Implementation

- Add the Gmail "Read Messages" node.
 - Connect it as a Tool to the Email Agent.
 - Return email information to the AI Agent.
 - Ensure the AI can summarize email content when requested.
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Part 2: Reply Gmail Tool

Create a tool that allows the AI to reply to emails.

Functional Requirements

The AI should be able to handle requests such as:

- Reply to the latest email and thank the sender.
- Respond to Ali and tell him I will attend the meeting.
- Reply professionally to the unread email.

Implementation

- Add the Gmail "Reply to Message" node.
- Connect it as a Tool to the Email Agent.
- Configure the node to use the correct Message ID or Thread ID.
- Ensure the AI generates appropriate reply content.
- The reply should be sent through Gmail.

Part 3: Supervisor Agent Integration

Update the Supervisor Agent so that it can intelligently decide when to:

User Request	Tool/Workflow
Send an email	Email Workflow
Read emails	Gmail Read Tool
Reply to email	Gmail Reply Tool
Search information	SerpAPI Tool

The Supervisor Agent should automatically select the correct tool based on the user's request.

Deliverables

Submit the following items on the student portal:

Screenshots

1. Screenshot of the Supervisor Workflow
2. Screenshot of the Email Workflow
3. Screenshot of the Gmail Read Tool Configuration
4. Screenshot of the Gmail Reply Tool Configuration

Video Demonstration

Create a video (2–5 minutes) demonstrating your completed solution.

The video must include:

- Overview of the Supervisor Workflow
- Overview of the Email Workflow
- Demonstration of reading emails using the AI Agent
- Demonstration of replying to an email using the AI Agent
- Successful execution of the workflow

Video Submission Instructions

1. Record your screen while running the workflow.
2. Upload the video to YouTube.
3. Set the visibility to either:
 - Public
4. Copy the YouTube video link.
5. Submit the YouTube link in the portal.

Important

- Do not upload the video file directly to the portal.
 - Only YouTube links will be accepted.
 - The workflow execution must be clearly visible in the recording.
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Gmail Read & Reply Tool Example Input Prompts and Expected Outputs

Test Case 1: Read Latest Email

User Input

Read my latest email.

Expected Agent Response

I found your latest email.

From: Ahmed Khan

Subject: Project Meeting Tomorrow

Summary: Ahmed has scheduled a project meeting for tomorrow at 10:00 AM and requested confirmation of attendance.

Test Case 2: Show Unread Emails

User Input

Show my unread emails.

Expected Agent Response

You have 3 unread emails:

1. Subject: Project Meeting Tomorrow
From: Ahmed Khan
2. Subject: Internship Opportunity
From: HR Team
3. Subject: Assignment Submission Reminder
From: Course Instructor

Test Case 3: Summarize Latest Email

User Input

Summarize my newest email.

Expected Agent Response

Summary:

The sender is inviting you to attend a project meeting scheduled for tomorrow. They have shared the agenda and requested confirmation of your availability.

Test Case 4: Reply to Latest Email

User Input

Reply to the latest email and thank the sender.

Expected Agent Response

Reply sent successfully.

Message:

Thank you for your email. I appreciate the information and look forward to attending the meeting.

Best regards,
[Your Name]

Test Case 5: Read Then Reply

User Input

Read my latest email.

Expected Agent Response

Latest Email:

Subject: Project Meeting Tomorrow

Summary: Meeting scheduled for tomorrow at 10:00 AM. Confirmation requested.

User Input

Reply that I will attend the meeting.

Expected Agent Response

Reply sent successfully.

Message:

Thank you for the invitation. I confirm that I will attend the meeting tomorrow.

Best regards,
[Your Name]

Test Case 6: Sender-Specific Email

User Input

Read the latest email from Ahmed.

Expected Agent Response

I found an email from Ahmed.

Subject: Project Meeting Tomorrow

Summary: Ahmed is requesting confirmation for tomorrow's meeting.